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Applied Econometrics: Empirical Project

Impact of the income on happiness according to the social context

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Abstract

Most of welfare economics studies focus on the relation between utility and quantities of consumption goods. Our micro-economic approach is not common as we do not only focus on a quantitative analysis, we include the social dimension which is qualitative. The aim of this paper is to study the impact of an income increase on welfare. It focuses on the evaluation of the individual well-being : not too happy, pretty happy, very happy. This paper also shows how the welfare of a person is likely to change to an income increase depending on her social characteristics (age, gender, religion, *etc.*), we focused on some of them : marital status, labor force status and education. Increasing income leads to a better life evaluation until the income reached approximately $87,000. Throughout, in order to provide rigorous results, as the outcome is an ordered categorical variable we used an ordered logistic regression model.

Contents

1 Introduction **3**

2 Literature review **4**

3 Data description **5**

3.1 Source 5

3.2 Summary statistics 5

4 Econometric specification **6**

5 Empirical results **8**

5.1 Impact of the income on happiness 8

5.1.1 First approach 8

5.1.2 Nuanced effect of an income increase on welfare 9

5.2 Impact of the income on happiness according to the social context 10

5.2.1 Impact of the income on happiness according to the marital status 11

5.2.2 Impact of the income on happiness according to labor force status and education 12

6 Conclusion **13**

References **14**

Tables **16**

Figures **20**

Appendices **26**

1 Introduction

Money is what governs our interactions daily, it allows us to buy consumption goods. The most common way to earn money is to have an income. Income increases the welfare of an individual in a certain measure, just make yourself the following thought experiment : if you do not have money, you will not be able to buy some foods, to live in your own apartment *etc.*; hence you will be hungry and get cold, as a consequence you will be less happy compared to the situation in which you have an income. Increasing the social welfare is a relevant goal, everybody wants to be happy. Furthermore, this aim is also written in the Declaration of the Rights of Man and of Citizens (1793), article n°1 (translation) : "The aim of society is the common welfare".

We first want to check if an increase of income enhances welfare or not and we will analyze precisely this impact. Then we will measure this impact following the social context. So the problematic is the following : what is the impact of an income increase on welfare ? Does the social context affects this increase ? By "social context" we mean all the social interactions the individual can have, what social characteristics does he have : number of children, marital status, education, *etc.*

We can consider for instance the case of a poor person and a wealthy one (the others characteristics are equal for both), does an increase of income impact the same way the welfare of those individuals ? We can ask ourselves several questions[1] : does the impact of an income increase on welfare is the same between a married person and an single one ? Is it the same between a man and a woman ? Does an income increase has the same effect on welfare between a person who is graduated and a person who is low educated ? These questions will be covered in the section dedicated to the results.

[1]For the following questions we assume that all the characteristics between the individuals are the same, except for the characteristic to be analyzed, obviously.

2 Literature review

The two Nobel laureates in economics: Daniel Kahneman and Angus Deaton (2010) have worked on the relation between high income and differences with the life evaluation and the emotional well being. It has been found that high earners have better life satisfaction, however people's day-to-day emotional well-being increases until an annual income threshold of $75,000. This suggests that welfare takes into account the income and suggests as well there exists a limit above which increasing income has no more effect on happiness.

Concerning the unemployment, Winkelmann (1998) found no evidence of the impact of unemployment on well-being, contrary to Gielen and Van Ours (2012) who has found that when workers become unemployed their happiness drops substantially. We expect that people without a job can be less happy than people who works, depending on their social characteristics[2].

Powdthavee, Lekfuangfu and Wooden (2013) analyzed the effect of education on happiness, they found there is an effect of an extra year of education on income, and also an effect of income on subjective well-being, but they did not find the evidence of an extra year of education on life satisfaction and advice future researchers to use the degree quality. We expect that there is a link between degree and happiness and we suggest that the degree quality is a more appropriate approach than the number of years of education.

Morevover, Chris M. Herbst and John Ifcher (2016) found that parents are happier over time relative to non-parents in contradiction with Luca Stanca (2012) who has found that having children is negatively related to well-being. This disagreement can be explain by the age of the parents and also the family income.

Throughout this paper, we will compare our results with those of the authors presented in this section.

[2] age, number of children, marital status, *etc.*

3 Data description

3.1 Source

The data used in this empirical project was produced by the General Social Survey[3]. After transformation (*cf.* Appendices, Table 11), we used cross-section data on 2506 observations from residents of United States. The survey was conducted on adults from 18 to 89 years and in agreement with the analysis we want to conduct we chose specific variables (*cf.* Table 1).

3.2 Summary statistics

The dependent variable is called *welfare*, with levels "Not too happy" ($welfare = 0$), "Pretty happy" ($welfare = 1$), and "Very happy" ($welfare = 2$)[4]. Most observations (57.06%) are "Pretty happy", 28.45% claims to be "Very happy" and the least represented population feels "Not too happy" (14.49%) (*cf.* Table 2).

Descriptive statistics from the quantitative variables inform us that, the mean of *inc*, which represents the annual income, is around $33,112, the average person is 49 years old and the number of children (*child*) is around 2 in average (*cf.* Table 4). If we look at the average of these variables by level of happiness (*cf.* Table 5) we can see that the average income increases according to the level of happiness (from $18,238 for "Not too happy" to $40,574 for "Very happy") contrary to average number of children which decreases from 2.11 to 1.86. Concerning the age (*age*) it does not seem to be any real difference.

Moreover we chose six qualitative variables (*cf.* Table 3), and as we can see (*cf.* Table 6) the dataset contains a large proportion of religious people ($relig = 1$), 77% of the observations are religious and 62% have a job ($work = 1$). Nonetheless high educated people ($educ = 1$) and non-white people ($nonwhite = 1$) are under-represented with 31% and 26% respectively.

[3] `gssdataexplorer.norc.org`
[4] (*cf.* Table 1)

Finally what is interesting to analyze is the average of dummy variables by level of happiness (*cf.* Table 7). This table suggests that the mean of *nonwhite* is decreasing with happiness from 32% to 24% contrary to the average of *married* which is increasing from 19% to 58%. Moreover the representation of *educ*, *work* and *married*, seems to be higher for people who are happy[5].

4 Econometric specification

In our analysis, the dependent variable (*welfare*) is a categorical variable which can takes three different values (*cf.* Table 1), as the outcome is ordered and can take more than two values, an appropriate estimation method is the ordered logit regression model[6].

The welfare of an individual depends on the income and the other control variables we include in our model [7]. So the relation is the following :

$$welfare^* = inc\beta_1 + age\beta_2 + female\beta_3 + relig\beta_4 + married\beta_5 + educ\beta_6 + work\beta_7 + child\beta_8 + nonwhite\beta_9 + u \text{ (M1)}$$

We assume the error term u follows a standard logistic distribution[8]. Note that it is written $welfare^*$ instead of $welfare$. $welfare^*$ is a latent variable that we cannot observe directly but in order to compute the probability of being "Not too happy", "Pretty happy" or "Very happy" we have to use the latent variable and two cut points (or threshold parameters) : α_1 and α_2[9].

$welfare = 0$ if $welfare^* \leq \alpha_1$

$welfare = 1$ if $\alpha_1 < welfare^* \leq \alpha_2$

$welfare = 2$ if $welfare^* > \alpha_2$

[5] *E.g.*, among those who stated to be "Very happy", 38% were high-educated, 58% were married, 65% were working, these percentages are lower for people who stated to be "Not too happy" or "Pretty happy".

[6] A multinomial logit regression model could have been used but this latter does not take into account that the outcome is ordered, as a consequence the ordered logit regression model is more appropriate in this case as an individual is happier when he answers he is "Very happy" rather than "Pretty happy", and he is obviously happier when he answers he is "Pretty happy" rather than "Not too happy"

[7] (*cf.* Table 1)

[8] u is a continuously distributed variable independent of X and the distribution of u is symmetric about zero.

[9] α_1 and α_2 are estimated by maximum likelihood (*cf.* Appendices, Maximum Likelihood).

We denote all the control variables by X and the corresponding parameters by β so the latent variable can be written as : $welfare^* = X\beta + u$.

E.g. the probability of being "Not too happy" can be computed as followed using the logit function[10] :

$$\mathbb{P}[welfare = 0|X] = \mathbb{P}[welfare^* \leq \alpha_1|X] = \mathbb{P}[u \leq \alpha_1 - X\beta|x] = \Lambda(\alpha_1 - X\beta)$$

The probability of being pretty happy and very happy are computed the same way and here are the corresponding formulas :

$$\mathbb{P}[welfare = 1|X] = \Lambda(\alpha_2 - X\beta) - \Lambda(\alpha_1 - X\beta)$$

$$\mathbb{P}[welfare = 2|X] = \Lambda(X\beta - \alpha_2)$$

As written in the literature review section, Kahneman and Deaton (2010) have found that people are no longer happier above an annual income of 75,000$, so we expect some non-linearity (like a concave function) caused by the corresponding variable *inc*, adding this variable to the square in the regression model is a solution to see if there is any value of *inc* after which the probability of being "Very happy" no longer increases. For this reason we added a second model in our analysis :

$$welfare^* = inc\zeta_1 + inc^2\zeta_2 + female\zeta_3 + nonwhite\zeta_4 + age\zeta_5 + educ\zeta_6 + work\zeta_7 + child\zeta_8 + married\zeta_9 + relig\zeta_{10} + \epsilon \quad \text{(M2)}$$

For (M2) we denote all the control variables by X and the corresponding parameters by ζ so the latent variable can be written as : $welfare^* = X\zeta + \epsilon$

We assume the error term ϵ follows a standard logistic distribution[11].

The probabilities are computed the same way :

$$\mathbb{P}[welfare = 0|X] = \Lambda(\alpha_1 - X\zeta)$$

[10] $\Lambda(.)$ is the logit function : $\Lambda(z) = \dfrac{e^z}{1+e^z}$

[11] ϵ is a continuously distributed variable independent of X and the distribution of ϵ is symmetric about zero.

$$\mathbb{P}[welfare = 1|X] = \Lambda(\alpha_2 - X\zeta) - \Lambda(\alpha_1 - X\zeta)$$

$$\mathbb{P}[welfare = 2|X] = \Lambda(X\zeta - \alpha_2)$$

5 Empirical results

5.1 Impact of the income on happiness

5.1.1 First approach

Does an income increase have an impact on happiness ? From (M1)[12], the estimated threshold parameters $(\widehat{\alpha_1}, \widehat{\alpha_2})$ are significant at the 1% level. $\widehat{\beta_1} = 0.06$ is positive and significant at the 1% level, so we expect *inc* to have a positive impact on the probability of being "Very happy" and a negative impact on the probability of being "Not too happy", the effect on the probability of being "Pretty happy" is more ambiguous[13]. However the value of the coefficient estimate is not interpretable, we cannot tell, as in an Ordinary Least Square (OLS) regression: *ceteris paribus*, the effect of the increase of one unit of *inc* on the predicted response probabilities is just the coefficient estimate associated to *inc* because in our case, the derivative of the response probabilities of $welfare = j|X, j \in \{0, 1, 2\}$ with respect to an explanatory variable takes into account the value taken by all the explanatory variables.

In order to know the partial effect of a variable on $welfare = j\ |X,\ j \in \{0, 1, 2\}$, marginal effects have to be computed. To provide representative results, marginal effects will be computed at the average point[14]: this particular "individual" is a low educated white woman of 49 years old who works and has 2 children, she earns \$33,112 per year, she is religious and single. At the average point, *ceteris paribus*, the partial effect of an increase of *inc* of one unit

[12] *cf.* Table 8

[13] *cf.* Appendices to see the formulas of the marginal effects, the sign of β_k determines the sign of the marginal effects on the probability of being "Not too happy" ($welfare = 0$) and on the probability of being "Very happy" ($welfare = 2$). The effect on the probability of being "Pretty happy" ($welfare = 1$) is determined by the sign of $\lambda(\alpha_1 - X\beta) - \lambda(\alpha_2 - X\beta)$, *cf.* Wooldridge (2010)

[14] *cf.* Table 9

($10,000) increases the predicted probability of being "Very happy" of 1.2 percentage point, decreases the predicted probability of being "Pretty happy" and "Not too happy" of 0.5 and 0.7 percentage point respectively. For more visual results, we can observe at the average point, the estimated response probabilities computed at each value of inc between 0 and 15 [15]. We observe that, as inc increases, the estimated response probabilities of $welfare = 0$ or $1|X$ decreases, and the estimated response probability of $welfare = 2|X$ increases (*cf.* Figure 1 & 2). Furthermore the gradient of the predicted response probability of $welfare = 0$ or $2|X$ remains approximately constant as inc increases, this is not the case for the predicated response probability of $welfare = 1|X$, for which the gradient decreases as inc increases. Graphically we conclude that the more income you have, the happier you will be. This idea can obviously be nuanced.

5.1.2 Nuanced effect of an income increase on welfare

Remember that, Kahneman and Deaton (2010), found that subjective well-being does not continue to increase as soon as the income reaches a certain value. Results (*cf.* Table 8) from (M2) inform us that, the threshold parameters are significant at the 1% level, $\widehat{\zeta_1}$ is positive and significant at the 1% level and $\widehat{\beta_1} < \widehat{\zeta_1} \Leftrightarrow 0.060 < 0.279$, it implies that (M1) underestimates the effect of inc. Indeed, at the average point, if we use (M1), the estimated marginal effect of inc on $welfare = 2|X$ is about 1.2 percentage point, whereas using (M2) leads the estimated marginal effect of inc to be 3.3 percentage points, nearly three times more[16]. $\widehat{\zeta_2}$ is negative and significant which implies that above a certain value taken by inc, the predicted response probability of $welfare = 2|X$ can decrease. We can look at the estimated marginal effect[17] of inc on $welfare = 2|X$ to verify that : $\frac{\partial \mathbb{P}[welfare = 2|X]}{\partial inc}(\widehat{\zeta}) = (\widehat{\zeta_1} + 2\widehat{\zeta_2} inc)\lambda(X\widehat{\zeta} - \widehat{\alpha_2}) = (0.279 - 0.032inc)\lambda(X\widehat{\zeta} - 2.277)$. As $\lambda(X\widehat{\zeta} - 2.277)$ is always positive, simply find the value

[15] the maximum value taken by the income variable in the dataset from GSS is about $131,677 ($inc = 13.17$), so we took $inc = 15$ as the upper bound, *cf.* Table 4.

[16] cf. Table 9 and 10 to compare estimated marginal effects between (M1) and (M2)

[17] *cf.* Appendices, Computation of marginal effects

of inc, let's call it inc^*, such that : $0.279 = 0.032inc^*$, in order to maximize the predicted response probability of being "Very happy". $inc^* = 8.7187$, it means that the predicted response probability of being "Very happy" increases until the income reaches $87,181 and then start to decrease a little (*cf.* Figure 3 & 4 for more visual results[18]). This value[19] is pretty similar but no equal to the value found by Kahneman and Deaton (2010) which is $75,000[20]. inc^* is the value that maximizes the predicted response probability of $welfare = 2|X$, $\forall X$. Thereby an income increase will not lead to the same increase of happiness whether the individual already has a high income or not, answering the question asked in the introduction.

5.2 Impact of the income on happiness according to the social context

From now on, we will use the results from (M2), as this latter provides more realistic results than (M1). Concerning the parameters estimates of the control variables representing the social context : the coefficient estimates associated to *married*, *educ*, and *work* are significant[21] and positive, so we expect a positive impact of these variables on happiness. *Ceteris paribus*, the marginal effect when $married = 1$ on the probability of being "Very happy" at the average point has to be considered as it in increased this response probability of 13.7 percentage points. *Ceteris paribus*, the estimated marginal effect when $work = 1$ on the probability of being "Very happy" at the average point increased this response probability of 4.4 percentage points, we found the same result than Gielen and Van Ours (2012), but this effect on happiness is minor. The coefficient estimates associated to $relig$ and *child* are both significant at the 5% level. *Ceteris paribus*, at the average point, the partial effect of having one more child on the response probability of $welfare = 2|X$ decreases this latter by 1.3 percentage point, this result

[18]We did not set inc to values greater than 15 ($150,000) for two reasons : it is unrealistic and the individual is likely to be unhappy for inc tending to $+\infty$.

[19]Comparatively, it is also the value for which the predicted response probability of $welfare = 0|X$ is the lowest.

[20]the data used by Kahneman and Deaton (2010) was not the same as they used data from 2009, however they also used data from United States of America.

[21]significant at the 1%, 10% and 5% level respectively.

has to be considered if the individual has several children, in agreement with Stanca (2012). Graphically we observe that the more children an individual have, the bigger is the shift of the curve representing the probability of being "Very happy", compared to not having children (*cf.* Figure 12).

To answer the question asked in the introduction, regarding the difference of the impact of an income increase on happiness between a man and a woman, it seems that there is no difference as the coefficient estimate associated to $female$ is not significant. It also holds for the coefficient associated to *nonwhite* which is not significant.

5.2.1 Impact of the income on happiness according to the marital status

At the average point we computed the predicted probabilities of the response probabilities of $welfare = j|X$ according to the level of income, whether the individual is married or not (*cf.* Figure 5 & 6). Graphically, we observe that the gap of the probability of being "Very happy" between a married person and a single one is not the same whether the individual has a high income or not. Therefore, it can be interesting to compute the difference of probability of $welfare = 2|X$ after an income increase between a married individual and a single one. *E.g.*, this difference is about 14.21 percentage points between a single individual and a married one earning \$26,000 a year[22], between a single individual and a married one earning \$87,000[23] a year it is about 16.77 percentage points. It means that: consider a person at the average point but earning \$26,000 a year, and being single, if this individual has her income changing from \$26,000 to \$87,000, her probability of being "Very happy" will obviously increase but not as much than the same individual being married. This difference[24] is about 2.56 percentage points. The interpretation is that a married person will interpret an income increase better than a single one, *ceteris paribus*. In other terms, being married will lead to a bigger expansion

[22] the rounded median annual income from the dataset we used. The median annual income is about \$25,740
[23] the rounded annual income that maximizes the response probability of $welfare = 2|X$ found in 5.1.2
[24] 0.1677 - 0.1421 = 0.0256

of happiness as a consequence of an income increase, than being single.

5.2.2 Impact of the income on happiness according to labor force status and education

At the average point we computed the predicted probabilities of the response probabilities of $welfare = j|X$ according to the level of income, whether the individual is graduated or not (*cf.* Figure 7 and 8) and whether the individual is working or not (*cf.* Figure 9 and 10). As in the previous part, graphically we observe that the gap[25] between an high-educated person and a low-educated one (same observation for persons who are working-unemployed) is not the same whether the individual has a high income or not. It can be relevant to compute difference of the predicted response probability of $welfare = 2|X$ after an income increase between an individual who is graduated and a person who is low educated, and also between a person who is working and a person who is unemployed. We found the same result as in the previous subsection[26], those who are graduated ($educ = 1$) are likely to interpret an income increase better than those who are low educated ($educ = 0$). Again, between a graduated person and a low educated, when the income changes from \$26,000 to \$87,000, here the difference is about 0.9 percentage point (which is minor), *ceteris paribus*.

Concerning the difference of probability of being "Very happy" between a person who works and a person unemployed when the income changes from \$26,000 to \$87,000, this difference is about 1.25 percentage points which is also minor. But both variables can be taken in account (*cf.* Figure 11). Graphically we check that the gap between the blue curve[27] and the red curve[28] is bigger for high-educated and working people than for low-educated and unemployed ones.

[25] for the probability of being "Very happy"
[26] *cf.* 5.2.1
[27] estimated probability of $welfare = 2|X$
[28] estimated probability of $welfare = 0|X$

6 Conclusion

To summarize, we have seen the impact of an income increase on happiness on several points of view. The first results inform us that an income increase does not affect happiness in the same measure whether the individual is poor or wealthy. Moreover being a man or a woman does not impact the effect of an income increase on happiness.

One of the most relevant result is that, according to the second model (M2), there exists a level of income that maximizes happiness[29], this value is around $87,000. Nearly the same result as Kahneman and Deaton (2010) who found this value around $75,000. We know there exist wealth inequalities like the fact[30] that "the richest 1% have more wealth than the rest of the world combined", as people are no longer happier above a certain value of income, the accumulation of wealth can be questioned.

About policy implications, consider a policy maker who wants to maximize the social welfare, he may take decisions like taxing more people earning $87,000 as above this value people are not happier and give this amount through social support for those who earn less in order to enhance their happiness. He can also encourage professional training to allow people to find a job and invest in education in order to give them chances to be highly educated.

To conclude, we first suggest to other researchers to use data in other countries to compare results across world regions. Furthermore, to use data with as much of informations as possible to extend this thematic (about individual health, work stress, leisure time, *etc.*). Including more emotional characteristics in the analysis in order to study the impact of happiness is a huge goal as we all want to maximize social welfare. For instance about work stress, a happy worker is more productive than someone who is not : everybody is benefiting from social welfare.

[29] probability of being "Very happy"
[30] *cf.* Hardoon; Fuentes-Nieva and Ayele (2016)

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Tables

Variable	Type	Description
welfare	dependent	indicates the level of happiness of an individual*
inc	independent	annual income in thousand of dollars
age	independent	age in years
female	independent	= 1 if the individual is a woman, 0 otherwise
relig	independent	= 1 if the individual is religious, 0 otherwise
married	independent	= 1 if the individual is married, 0 otherwise
educ	independent	= 1 if the individual is graduated, 0 otherwise
work	independent	= 1 if the individual has a job, 0 otherwise
child	independent	number of children
nonwhite	independent	= 1 if the individual is white of skin, 0 otherwise

* : = 0 (Not too happy), = 1 (Pretty happy), = 2 (Very happy)

Table 1: Variable descriptions

welfare	**Observations**	**Frequency**
Not too happy	363	0.1449
Pretty happy	1430	0.5706
Very happy	713	0.2845

Table 2: Number of observations for each happiness level

Variable	**Label**	**Observations**
female	(= 1) female	1370
	(= 0) male	1136
nonwhite	(= 1) white	1861
	(= 0) non-white	645
educ	(= 1) graduated	787
	(= 0) not graduated	1719
work	(= 1) working	1546
	(= 0) unemployed	960
married	(= 1) married	1077
	(= 0) single	1429
relig	(= 1) religious	1936
	(= 0) atheist	570

Table 3: Dependent qualitative variables observations

Variable	**Min**	**Avg**	**Max**	**S.E**
inc	0.0234	3.3112	13.1677	3.1641
age	18	48.62	89	17.37
child	0	1.8	9	1.6

Table 4: Summary of dependent quantitative variables

welfare	Avg *inc*	Avg *age*	Avg *child*
Not too happy	1.824	49.51	2.1
Pretty happy	3.317	47.96	1.8
Very happy	4.057	49.50	1.9

Table 5: Average of quantitative variables by welfare

Variable	**Min**	**Avg**	**Max**
female	0	0.55	1
nonwhite	0	0.26	1
educ	0	0.31	1
work	0	0.62	1
married	0	0.43	1
relig	0	0.77	1

Table 6: Summary statistics of dummy variables

welfare	*female*	*nonwhite*	*educ*	*work*	*married*	*relig*
Not too happy	0.58	0.32	0.16	0.45	0.19	0.75
Pretty happy	0.54	0.25	0.32	0.64	0.42	0.76
Very happy	0.54	0.24	0.38	0.65	0.58	0.8

Table 7: Average of dummy variables by welfare

Dependent variable : *welfare*		
Explanatory variable	**M1**	**M2**
inc	0.060^{***} (0.015)	0.279^{***} (0.045)
inc^2		-0.016^{***} (0.003)
age	0.003 (0.003)	0.002 (0.003)
female	-0.012 (0.081)	-0.005 (0.081)
relig	0.190^{*} (0.097)	0.195^{**} (0.097)
married	0.814^{***} (0.091)	0.692^{***} (0.095)
educ	0.242^{***} (0.093)	0.179^{*} (0.094)
work	0.311^{***} (0.092)	0.232^{**} (0.093)
child	-0.068^{**} (0.027)	-0.061^{**} (0.027)
nonwhite	-0.009 (0.095)	0.031 (0.095)
α_1	-0.893^{***} (0.179)	-0.648^{***} (0.185)
α_2	2.002^{***} (0.183)	2.277^{***} (0.192)
N	2506	2506
LR Khi2	222.075	248.652
Prob > χ^2	0	0
Pseudo R-Squared	0.046	0.051

*** : significant at the 1% level, **: significant at the 5% level, *: significant at the 10% level

Table 8: Ordinal logistic regression coefficient

Dependent variable : *welfare*			
Explanatory variable	Not too happy	Pretty happy	Very Happy
inc	-0.007	-0.005	0.012
relig	-0.023	-0.014	0.036
married	-0.090	-0.072	0.162
educ	-0.027	-0.021	0.048
work	-0.037	-0.023	0.060
child	0.008	0.006	-0.013

Table 9: Estimated marginal effects at the average point (M1)

Dependent variable : $welfare$			
Explanatory variable	Not too happy	Pretty happy	Very Happy
inc	-0.020	-0.014	0.033
relig	-0.023	-0.014	0.037
married	-0.076	-0.061	0.137
educ	-0.020	-0.016	0.035
work	-0.027	-0.018	0.044
child	0.007	0.005	-0.012

Table 10: Estimated marginal effects at the average point (M2)

Figures

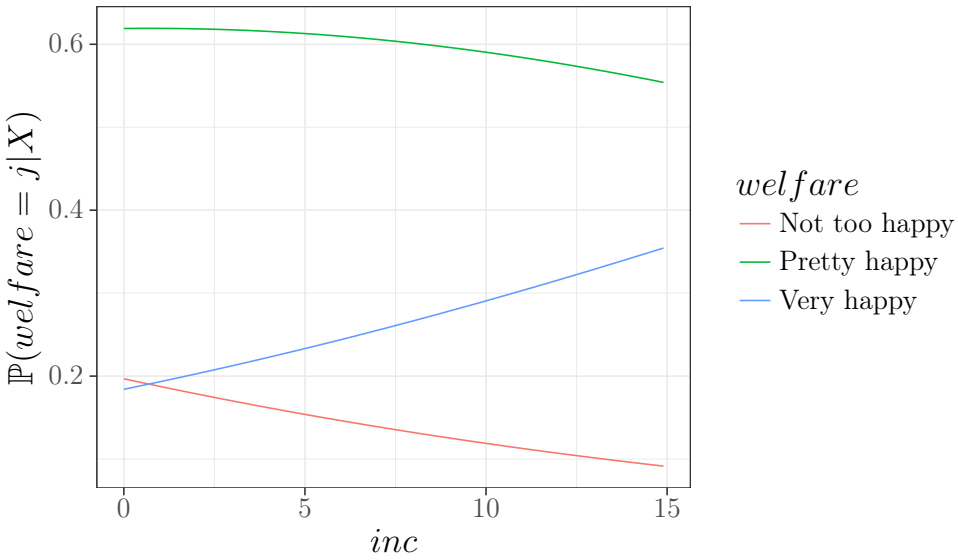

Figure 1: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point using model (M1) (1)

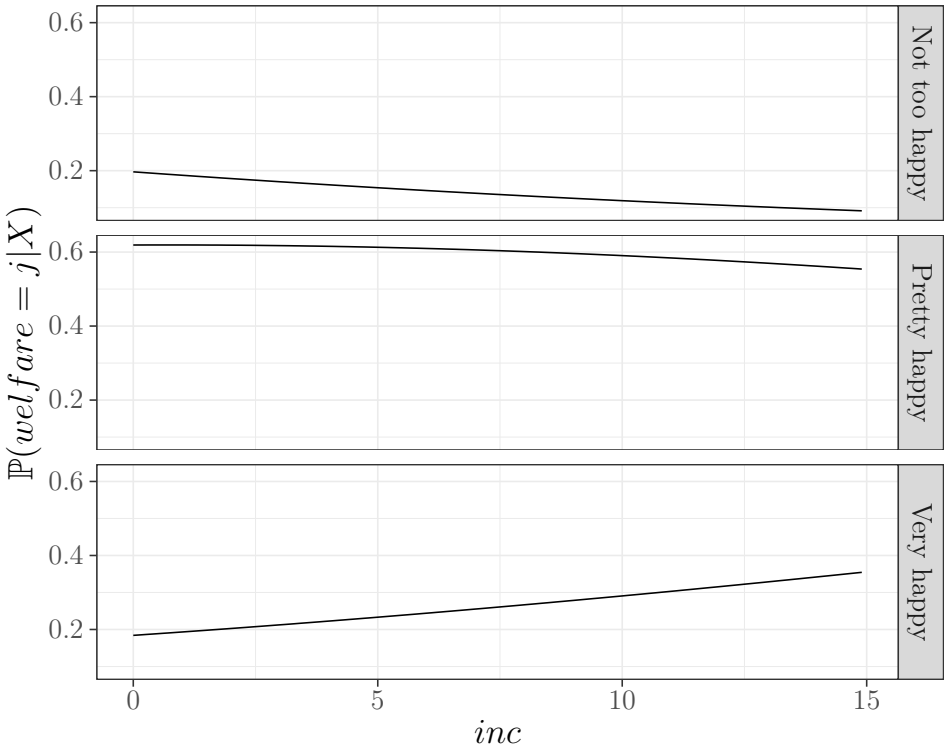

Figure 2: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point using model (M1) (2)

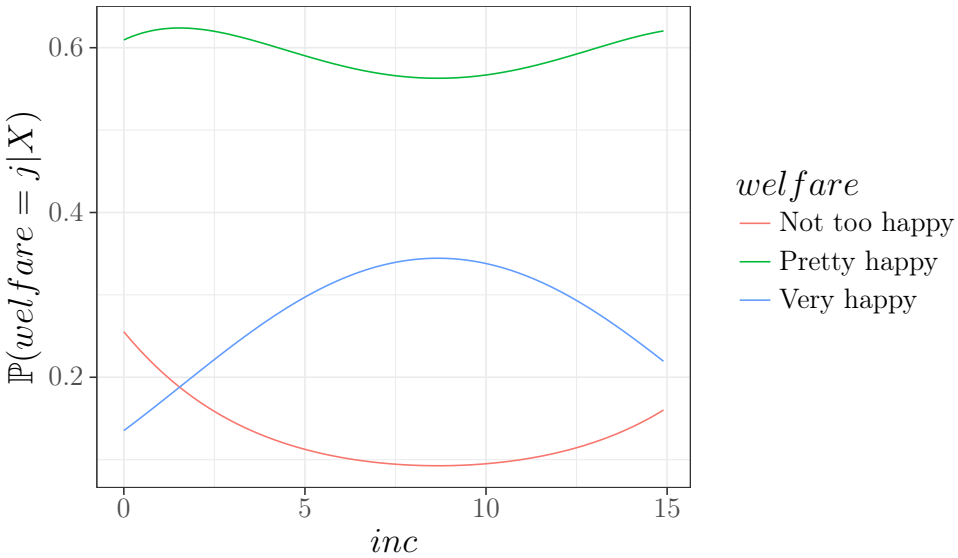

Figure 3: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point using model (M2) (1)

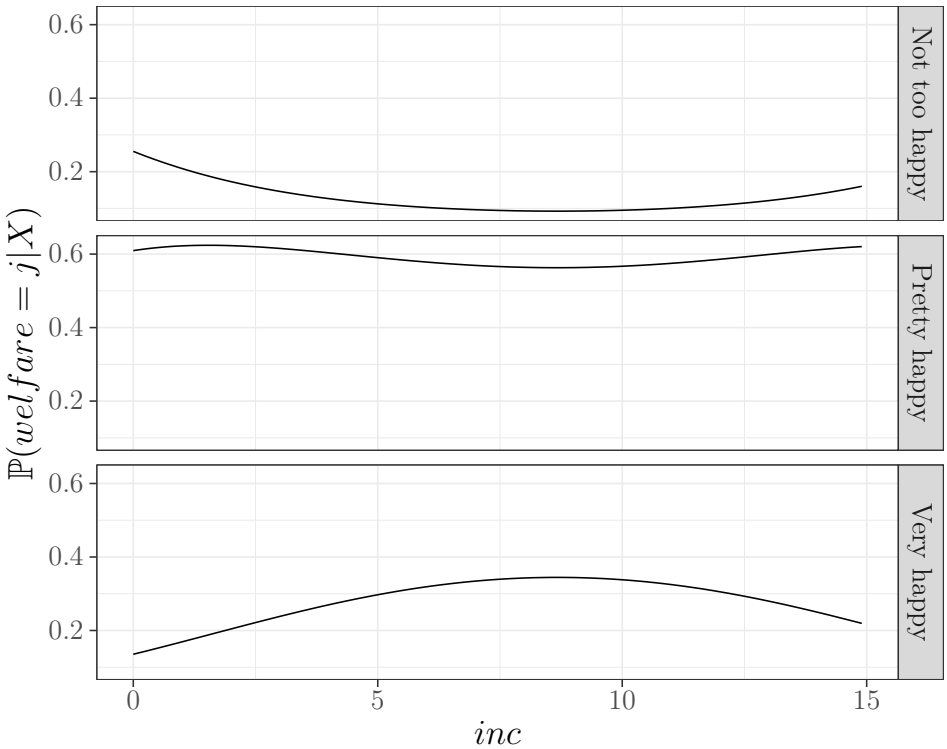

Figure 4: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point using model (M2) (2)

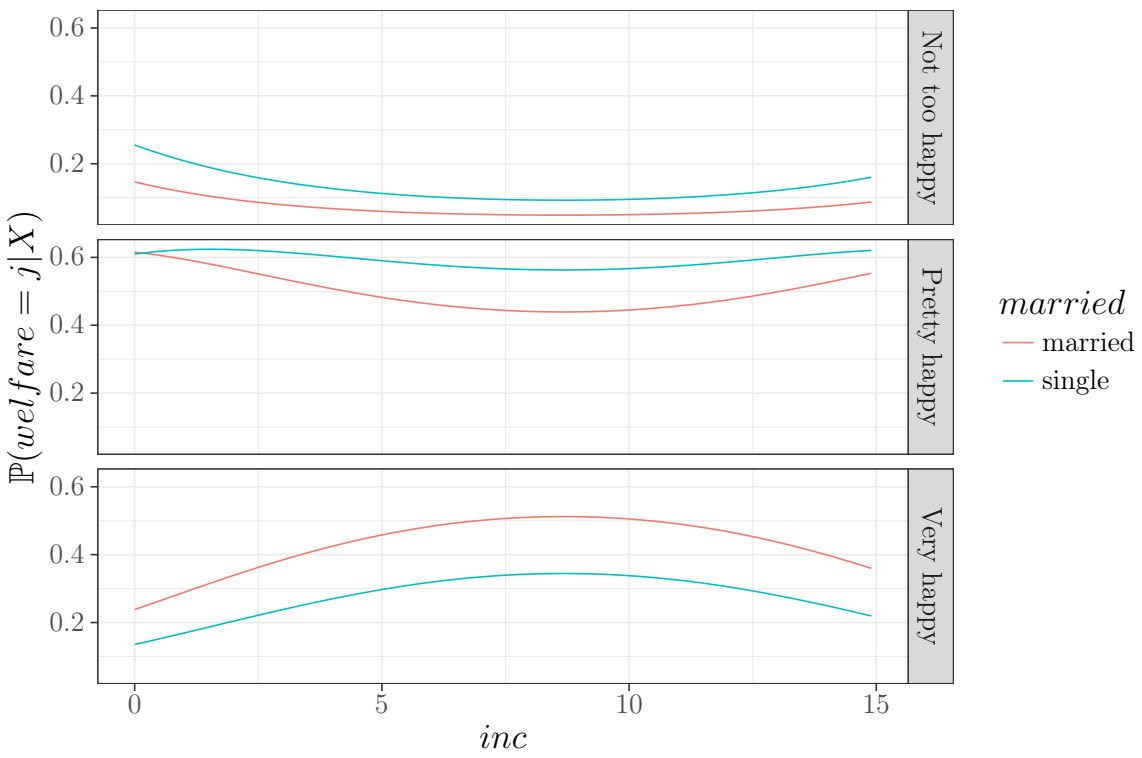

Figure 5: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual is married or not, using model (M2)

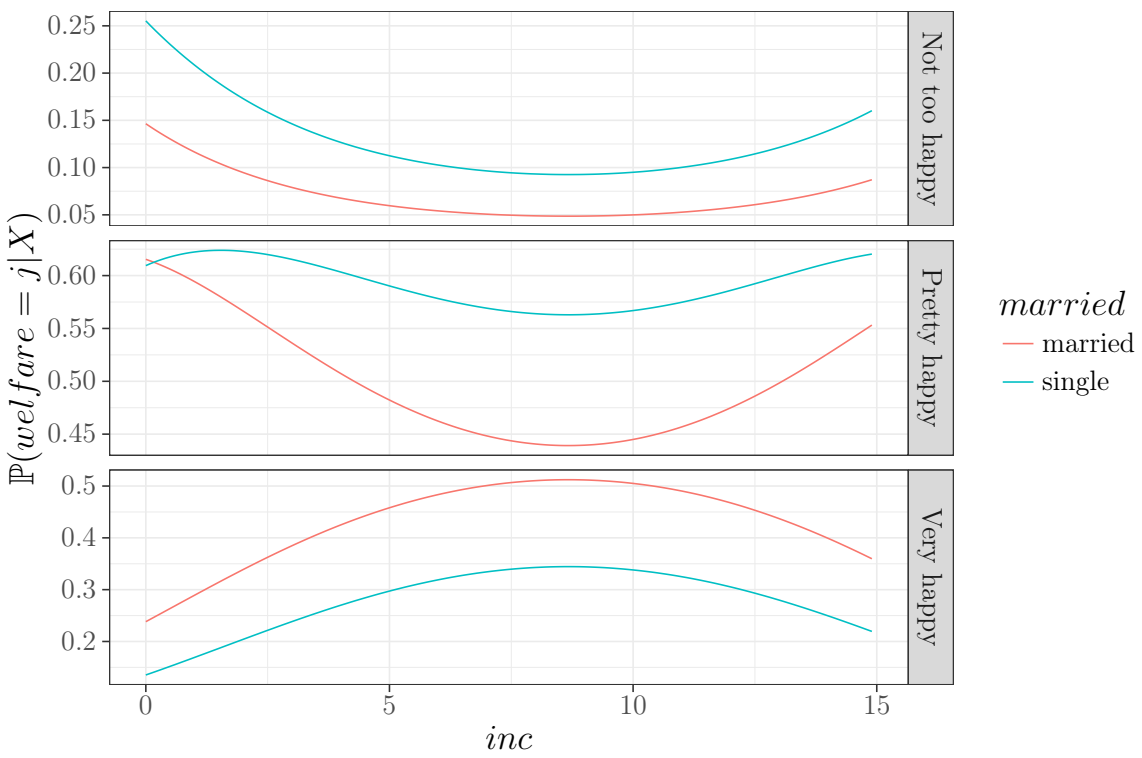

Figure 6: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual is married or not, using model (M2) (scale zoomed)

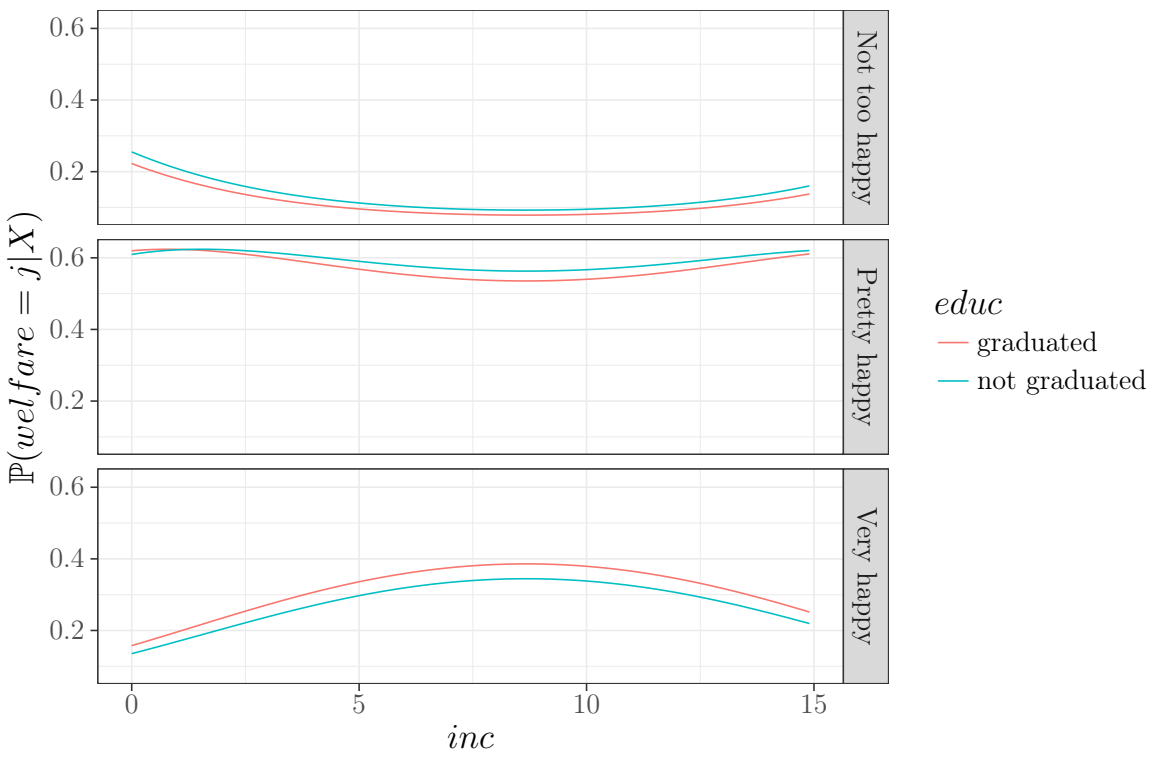

Figure 7: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual is graduated or not, using model (M2)

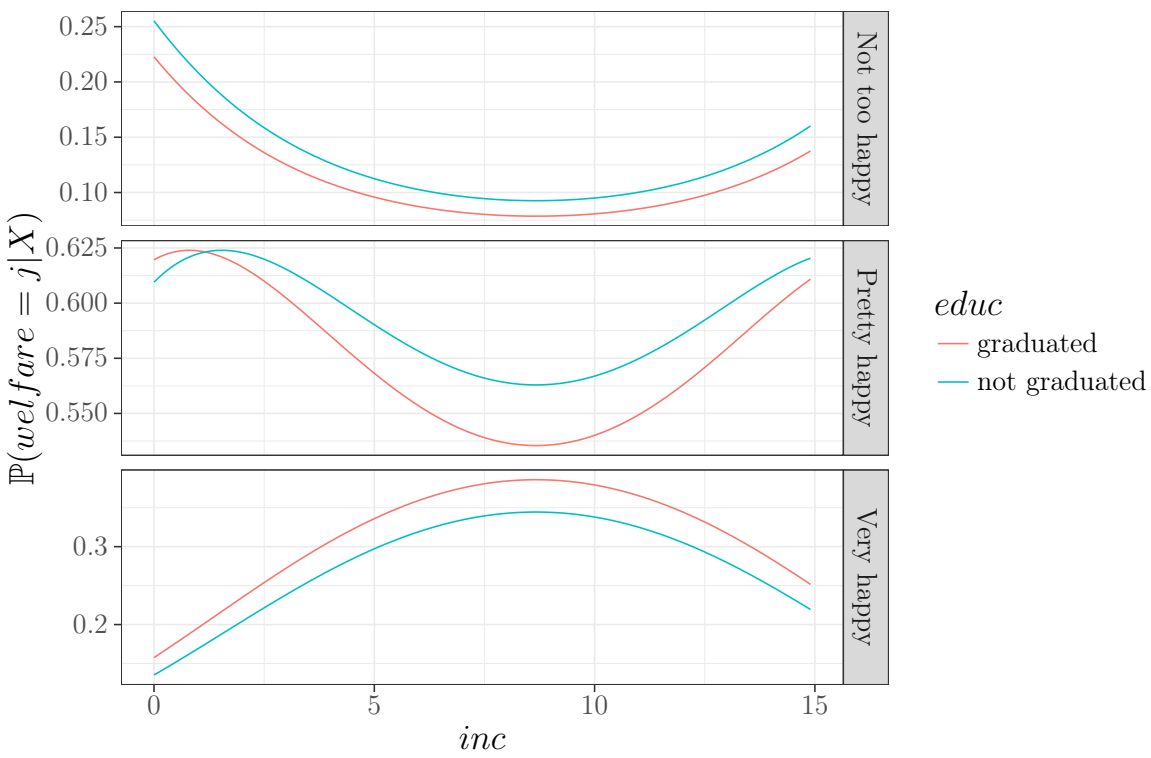

Figure 8: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual is graduated or not, using model (M2) (scale zoomed)

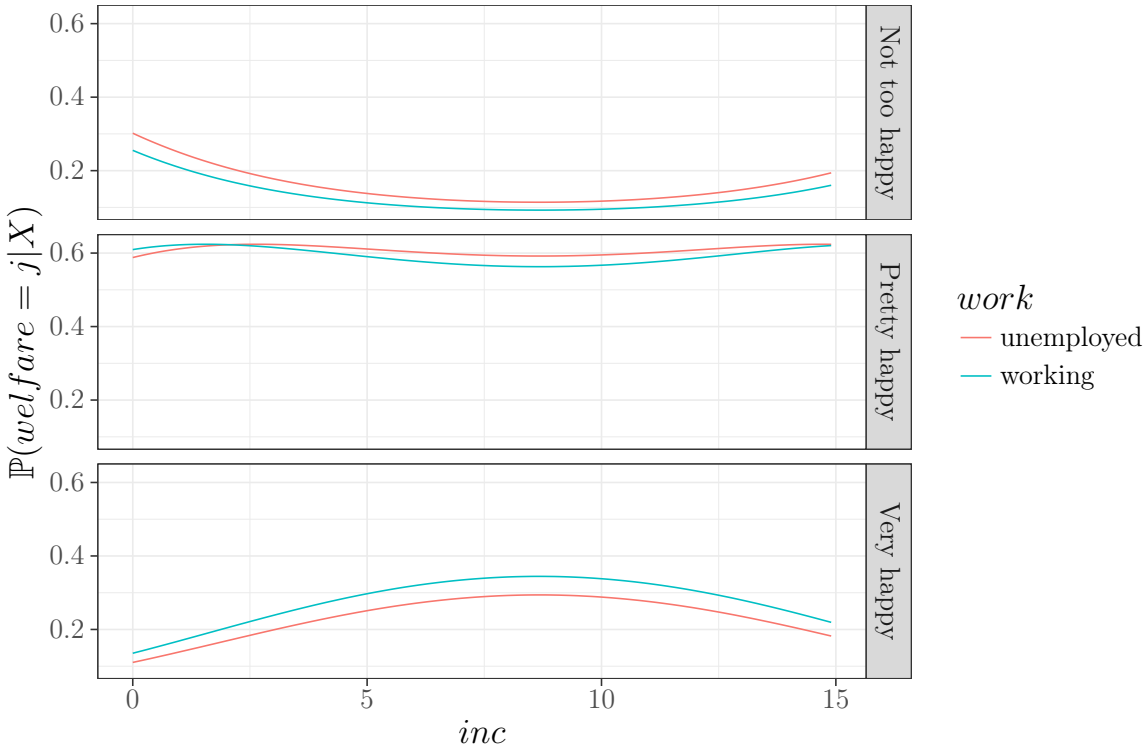

Figure 9: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual is working or not, using model (M2)

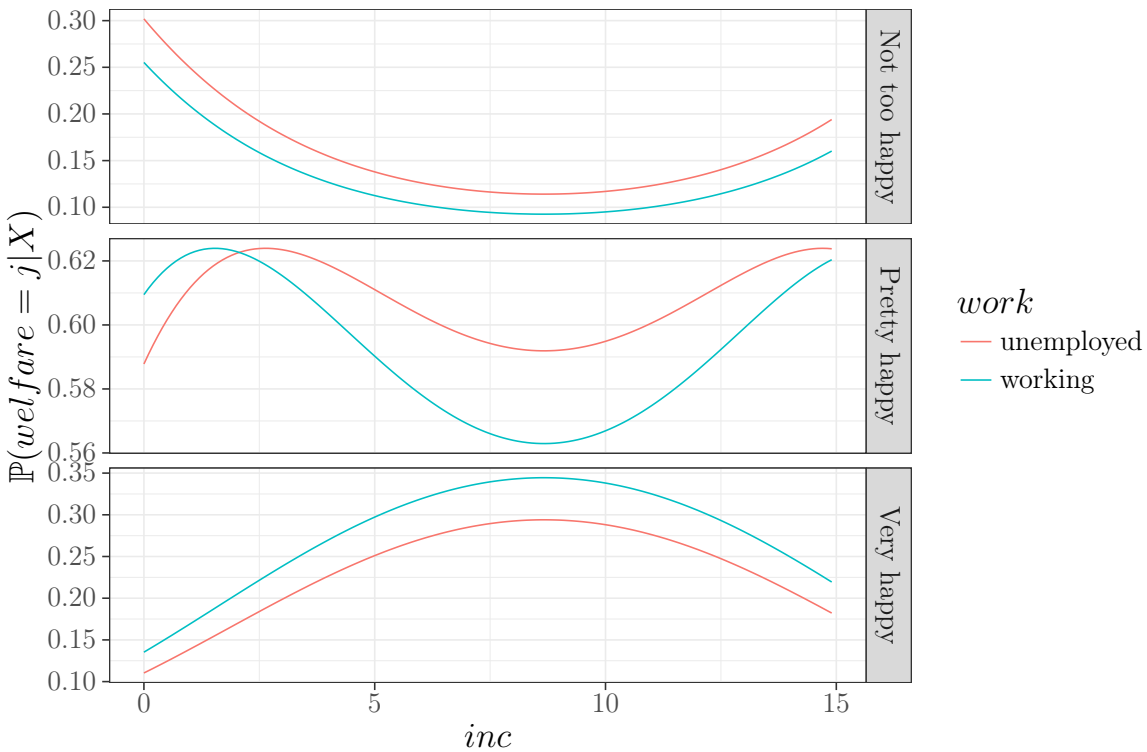

Figure 10: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual is working or not, using model (M2) (scale zoomed)

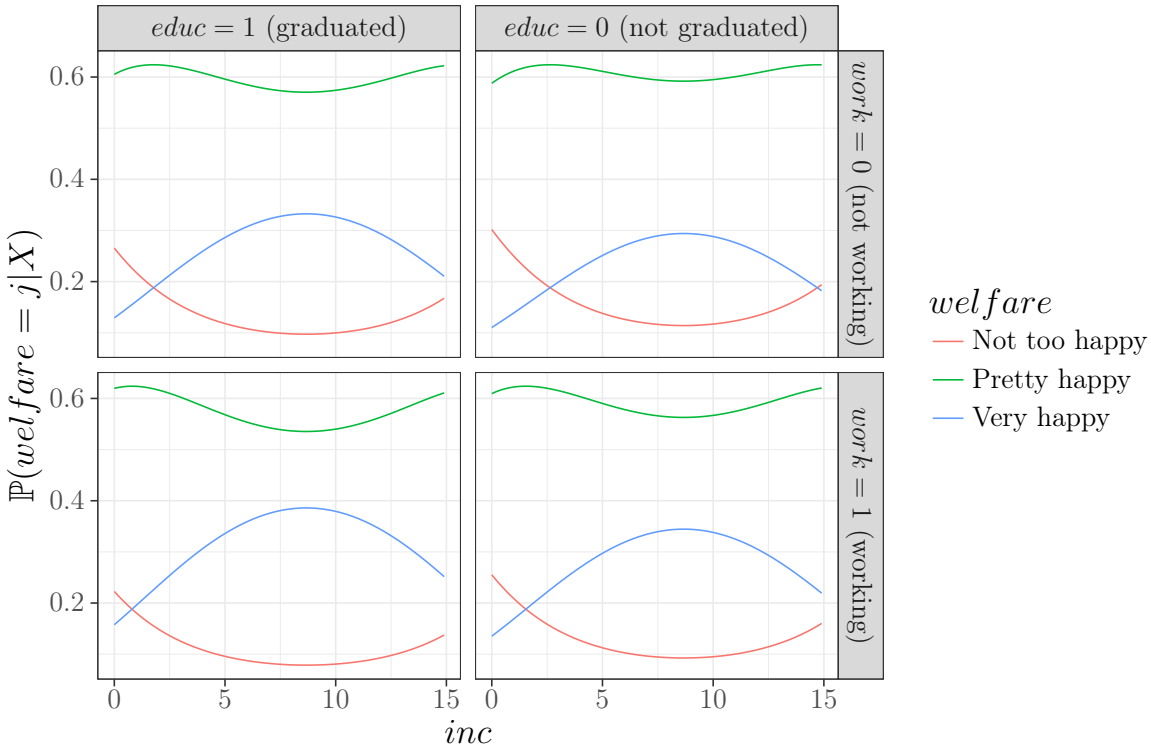

Figure 11: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual is working/or not and is graduated/or not, using model (M2)

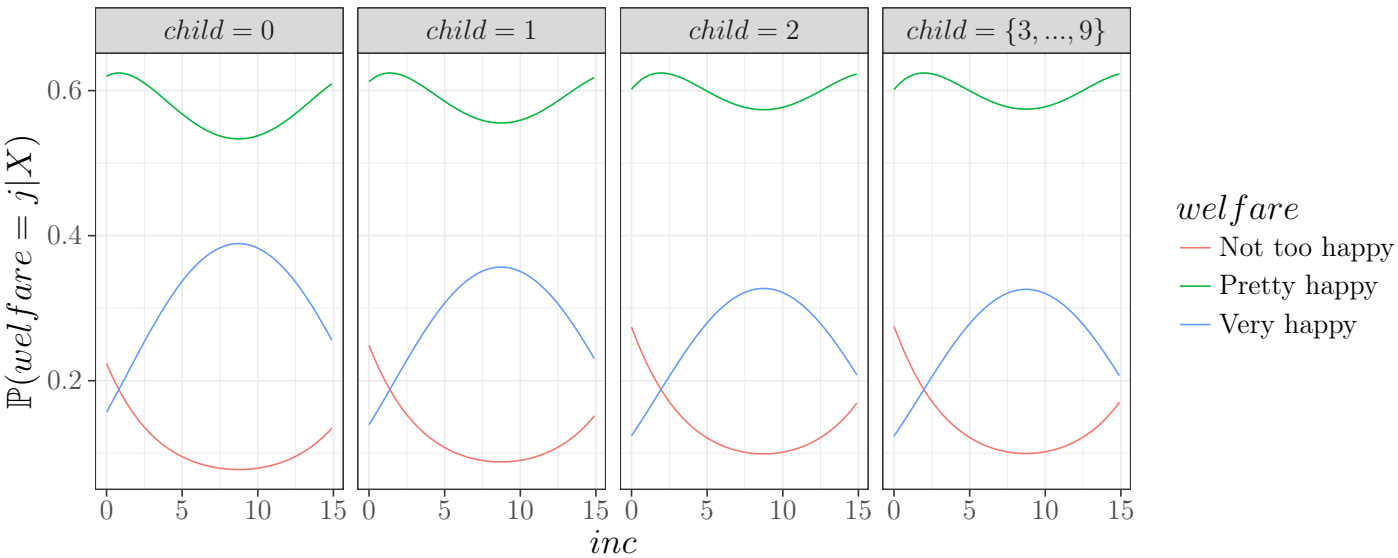

Figure 12: Predicted probabilities of $welfare = j|X$ according to the level of income (inc) at the average point, whether the individual has 0, 1, 2 or 3 & more children, using (M2)

Appendices

Survey question	**Answer**	**Transformation**
Taken all together, how would you say things are these days – would you say that you are very happy, pretty happy, or not too happy ?	1 Very happy 2 Pretty happy 3 Not too happy 8 Don't know 9 No answer 0 Not applicable	We only keep "Very Happy" ($welfare = 2$), "Pretty happy" ($welfare = 1$) and "Not too happy" ($welfare = 0$).
Family income on 1972-2016 surveys in constant dollars (base = 1986)	Family income	We divide the income by 10000 in order to have a more interpretable coefficient estimate.
Code respondent's sex	1 male 2 female	No change.
What race do you consider yourself ?	1 White 2 Black 3 Other	We gathered "Black" and "Other" as people who are non-white ($nonwhite = 1$), we keep "White" for white people ($nonwhite = 0$).
Respondent's age	Age	No change.

Respondent's degree	0 Lt high school 1 High school 2 Junior college 3 Bachelor 4 Graduate 8 Don't know 9 No answer	We deleted 8 and 9 and consider 3 & 4 as high education ($educ = 1$), otherwise, for 0, 1, 2, we consider those ones as low education ($educ = 0$).
Last week were you working full time, part time, going to school, keeping house, or what ?	1 Working full time 2 Working part time 3 Temp not working 4 Unemployed, laid off 5 Retired 6 School 7 Keeping house 8 Other 9 No answer	We deleted 8 & 9 and consider 1 and 2 as people who are working ($work = 1$). Otherwise, for 3, 4, 5, 6 and 7, we consider those ones as not working ($work = 0$)
How many children have you ever had ? Please count all that were born alive at any time (including any you had from a previous marriage)	Number of children	No change.

Are you currently married, widowed, divorced, separated, or have you never been married ?	1 Married 2 Widowed 3 Divorced 4 Separated 5 Never married 9 No answer	We deleted 9 and keep 2 as people who are married ($married = 1$) otherwise, for 2, 3, 4, and 5, we consider those ones as people who are not married ($married = 0$).
What is your religious preference ? Is it Protestant, Catholic, Jewish, some other religion, or no religion ?	1 Protestant 2 Catholic 3 Jewish 4 None 5 Other 6 Buddhism 7 Hinduism 8 Other eastern 9 Moslem/islam 10 Orthodox-christian 11 Christian 12 Native american 13 Inter-nondenominational 98 Don't know 99 No answer	We deleted 98 and 99, we consider 4 as people who are not religious ($relig = 0$) and ($relig = 1$) otherwise (for 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13).

Table 11: Survey questions

Computation of marginal effects

$$\frac{\partial \mathbb{P}[welfare = 0|X]}{\partial x_k} = \begin{cases} -\beta_k \lambda(\alpha_1 - X\beta) & \text{for (M1)} \\ -\zeta_k \lambda(\alpha_1 - X\zeta) & \text{for (M2)} \end{cases}$$

$$\frac{\partial \mathbb{P}[welfare = 1|X]}{\partial x_k} = \begin{cases} \beta_k [\lambda(\alpha_1 - X\beta) - \lambda(\alpha_2 - X\beta)] & \text{for (M1)} \\ \zeta_k [\lambda(\alpha_1 - X\zeta) - \lambda(\alpha_2 - X\zeta)] & \text{for (M2)} \end{cases}$$

$$\frac{\partial \mathbb{P}[welfare = 2|X]}{\partial x_k} = \begin{cases} \beta_k \lambda(X\beta - \alpha_2) & \text{for (M1)} \\ \zeta_k \lambda(X\zeta - \alpha_2) & \text{for (M2)} \end{cases}$$

Note that : $\lambda(z) = \dfrac{\partial \Lambda(z)}{\partial z} = \dfrac{\Lambda(z)}{1 + e^z}$

Exception : in (M2), inc^2 is present in $X\zeta$, therefore the computation of the marginal effect of inc on $welfare = j|X$ is different :

$$\frac{\partial \mathbb{P}[welfare = 0|X]}{\partial inc} = -(\zeta_1 + 2\zeta_2 inc)\lambda(\alpha_1 - X\zeta)$$

$$\frac{\partial \mathbb{P}[welfare = 1|X]}{\partial inc} = (\zeta_1 + 2\zeta_2 inc)[\lambda(\alpha_1 - X\zeta) - \lambda(\alpha_2 - X\zeta)]$$

$$\frac{\partial \mathbb{P}[welfare = 2|X]}{\partial inc} = (\zeta_1 + 2\zeta_2 inc)\lambda(X\zeta - \alpha_2)$$

Maximum Likelihood

$\forall i \in \{1, ..., 2506\}$, the log-likelihood function is :

(M1) : $l_i(\alpha_1, \alpha_2) = \mathbf{1}[welfare_i = 0] log[\Lambda(\alpha_1 - X_i\beta)] + \mathbf{1}[welfare_i = 1] log[\Lambda(\alpha_2 - X_i\beta) - \Lambda(\alpha_1 - X_i\beta)] + \mathbf{1}[welfare_i = 2] log[\Lambda(X_i\beta - \alpha_2)]$

(M2) : $l_i(\alpha_1, \alpha_2) = \mathbf{1}[welfare_i = 0] log[\Lambda(\alpha_1 - X_i\zeta)] + \mathbf{1}[welfare_i = 1] log[\Lambda(\alpha_2 - X_i\zeta) - \Lambda(\alpha_1 - X_i\zeta)] + \mathbf{1}[welfare_i = 2] log[\Lambda(X_i\zeta - \alpha_2)]$